Stability-Constrained Regime-Aware Forecasting for Heterogeneous Panel Time Series

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Abstract

Social science panel time series exhibit multicollinearity, latent unit heterogeneity, and regime-conditional sign-changing relationships. On V-Dem democratic development data, four of four selected covariates exhibit coefficient sign reversal across regimes, driving pooled additive predictors to cancel real signal - a diagnostic we establish before introducing any nonlinear model. We present a regime-aware additive forecasting architecture that recovers this signal under a verifiable contraction certificate. Our theoretical contribution is a rate-based contraction theorem decomposing the global Lipschitz constant of a soft-gated mixture predictor into per-regime, regime-disagreement, and dwell-time components, each empirically auditable on a trained model. On V-Dem democratic development data (151 countries, 1970–2025), the architecture recovers regime-conditional signal that pooled additive models cancel (paired Diebold–Mariano $p = 0.016$), with the recovery localized to the tails of the democracy index where a diagnostic shows largest covariate sign disagreement. The contraction certificate $L_{\text{eq,max}} < 1$ holds in $5/5$ seeds across all 9 audited configurations without explicit projection. A synthetic stress test with stronger regime contrasts shows the certificate is non-vacuous: spectral normalization on per-edge layers can fail to deliver contraction because the gating-driven term $L_{\phi,\text{eq}} \cdot M$ dominates, and a regime-disagreement penalty is required to recover the bound.

1 Introduction

Forecasting heterogeneous panel time series is a central problem in the social sciences [1]. These data exhibit three persistent challenges: high-dimensional multicollinearity among predictors, latent heterogeneity across units, and regime-dependent relationships that may vary in magnitude or even change sign across states [1; 2]. Classical linear forecasters (ARIMA, vector autoregression [3]) do not address these challenges jointly: their extensions either rely on strong parametric assumptions or scale poorly. Threshold and Markov-switching VAR [4; 5] represent regime structure but inherit linear within-regime dynamics. Recent neural autoregressive approaches [6] offer flexible nonlinear alternatives, but in recursive forecasting settings expressive nonlinear models can amplify small perturbations over time; the standard stability tool [7]- spectral normalization on per-edge layers - turns out to be insufficient for soft-gated mixture architectures, where the gating-driven contribution to the Lipschitz constant is independent of per-edge spectral norms and can dominate.

We present **PANAR-X**, a regime-aware additive forecasting architecture for heterogeneous panel time series with formal stability conditions and an empirical audit protocol. The architecture builds on NAVAR-style per-edge additive networks [6], originally developed for nonlinear Granger-causal structure discovery, and adapts them for panel forecasting in two ways: separating the endogenous lag channel from the predetermined covariate channel, and introducing per-unit intercepts. It further adds a regime-aware soft gating mechanism that captures latent regime structure without hard data

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partitioning, alongside a contraction analysis to ensure recursive forecast stability. We focus on forecasting a panel of democracy indicators from the V-Dem [8], treating them as country-year time series with heterogeneous, weak-signal dynamics.

Contributions. (i) We introduce a regime-aware additive forecasting architecture for panel data, together with a rate-based contraction theorem that decomposes the global Lipschitz constant into per-regime, regime-disagreement, and dwell-time components, each empirically auditable. (ii) We identify a failure mode of standard stability tools in soft-gated mixture architectures: on a synthetic panel with strong regime contrasts, spectral-norm projection of per-edge layers fails to ensure contraction because the gating term $L_{\phi,\mathrm{eq}} \cdot M$ drives the equilibrium-step Lipschitz constant above unity, even when per-regime $L_Y^{(r)}$ is controlled. A regime-disagreement penalty λ_M on M is therefore required to recover the bound; a 45-run sweep maps the resulting stability–accuracy Pareto frontier. (iii) On a democratic development dataset from V-Dem [8], we show that pooled additive models cancel regime-conditional, sign-changing relationships, whereas the proposed architecture recovers these signals under strict stability constraints (paired Diebold–Mariano $p = 0.016$ vs. pooled additive baseline; $L_{\mathrm{eq,max}} < 1$ in $5/5$ seeds across 9 audited configurations), with gains concentrated in the tails of the democracy index where covariate sign-disagreement is largest.

Scope. The forecasting tasks we study are intrinsically difficult: simple baselines such as random walks perform strongly, and absolute predictive gains are modest. Accordingly, our objective is not to maximize accuracy, but to show that regime-dependent structure can be modeled in a principled way without sacrificing stability.

The contraction theorem serves two roles. In synthetic settings with strong regime contrasts, it characterizes when expressive nonlinear forecasters would otherwise diverge under recursive rollout; the experiment shows the constraint is non-vacuous and identifies the training-time controls needed to restore contraction. In real-world panel data, such as V-Dem data [8], where heterogeneity, multicollinearity, and weak signals dominate, the constraint typically holds under standard training configurations; the audit then confirms safe recursive deployment.

2 Related work

Linear and regime-switching panel forecasting. Classical multivariate panel forecasting relies on vector autoregression with n^2p parameters in an unrestricted $\mathrm{VAR}(p)$ on n series; sparse and structured penalizations [3] address the high-dimensional case but do not impose state-dependent dynamics. The forecasting-with-panel-data literature documents the tension between pooled and per-unit estimators [1], neither of which represents regime-conditional sign-changing relationships without explicit regime structure. Markov-switching VAR (MS-VAR) [5; 9] introduces hidden-Markov regimes with linear within-regime dynamics; threshold VAR (T-VAR) [4] uses an observable threshold variable. Both inherit linear dynamics within regimes and provide stability conditions for individual regime matrices rather than for the switching system as a whole. Recent neural extensions [10] introduce nonlinear within-regime dynamics via deep state-space models but do not address verifiable contraction of the resulting switching system. Dynamical-systems modeling of social science panel data [11] similarly identifies regime structure (development traps) but does not provide a forecasting architecture with stability guarantees.

Neural additive autoregression and our positioning. Neural additive vector autoregression (NAVAR) [6] introduces per-edge nonlinear networks summed in an additive predictor, originally proposed and validated for nonlinear Granger-causal structure discovery on multivariate time series. NAVAR's forecasts were used as a score for Granger links rather than as a forecasting deliverable, and the architecture does not represent regime structure: per-edge functions are pooled across the state space, so sign-changing regime-conditional effects cancel under pooling. Recent work [12] on function-valued causal influence shows representationally that scalar summaries collapse such sign-changing relationships; this motivates our regime-conditional architectural extension. PANAR-X repurposes the NAVAR per-edge structure for forecasting, separates the endogenous lag channel from the predetermined covariate channel, adds per-unit intercepts for panel heterogeneity, and adds regime-aware soft gating with a contraction analysis that NAVAR, MS-VAR, and TVAR each lack.

Stability of switching systems and neural Lipschitz control. Our stability theorem specializes the switching-system average-dwell-time framework of [13] to soft-gated neural mixture architectures. Where the classical analysis bounds switched-system trajectories in terms of generic per-mode Lipschitz constants, our Theorem 1 expresses the dwell-time threshold α^* in terms of architecture-derived constants ($L_{\text{eq,max}}$, B_{trans}) that are individually computable on the trained model and individually controllable through training. Penalty-based enforcement follows spectral-norm regularization practice [7; 14], with the addition of a regime-disagreement penalty addressing the gating-driven contribution to the bound - a contribution we identify as essential because spectral-norm projection on per-edge layers alone can fail to deliver contraction (§7).

3 Architecture

PANAR-X uses NAVAR-style per-edge additive networks [6]: a separate small neural network for each lag-history-to-target edge, with the predictor formed as an additive sum of edge contributions. We adapt NAVAR for panel forecasting with three modeling choices: (i) per-unit intercepts to absorb panel-level heterogeneity; (ii) a separate channel for predetermined structural covariates (X), distinguished from the endogenous target-history channel (Y); and (iii) the regime-aware soft-gating extension and contraction certificate developed in this paper, which is the methodological contribution.

We work with three model variants of increasing expressiveness: **M1** (pooled additive, Y-channel only), **M1.5** (pooled additive, Y+X channels), and **M2** (regime-aware: a soft-gated mixture of R regime-specific M1.5 predictors). M2 reduces to M1.5 in the single-regime limit. The empirical sections compare M2 against M1 and M1.5 to isolate the contribution of regime structure (M2 vs. M1.5) from the covariate channel (M1.5 vs. M1) and from nonlinearity (M1 vs. AR(1)). The remainder of this section specifies M2; M1 and M1.5 are pooled specializations.

Panel setup. A panel of U units indexed by u observed at times $t = 0, \dots, T_u - 1$. The model receives lagged histories $\mathcal{H}_t^{Y,(u)} \in \mathbb{R}^{L \times K_Y}$ and $\mathcal{H}_t^{X,(u)} \in \mathbb{R}^{L \times K_X}$. The state space $\mathcal{Z} = \mathbb{R}^{L \times K_Y} \times \mathbb{R}^{L \times K_X}$ is taken compact, with the Frobenius norm on history matrices and the ℓ_2 operator norm on Jacobians.

Regime function. The regime indicator is a deterministic Lipschitz functional of state. Let $\psi : \mathcal{Z} \to \mathbb{R}$ be a Lipschitz scalar summary, in our applications $\psi(z) = z[L][0]$, the most recent value of the dominant endogenous factor (with $L_\psi = 1$). With fixed regime centers $\{c_r\}_{r=1}^R$ (tertile midpoints of ψ on training data), the regime weights are a tempered softmax

$$\pi_r(\psi) = \frac{\exp(-(\psi - c_r)^2/\tau)}{\sum_{r'} \exp(-(\psi - c_{r'})^2/\tau)}. \tag{1}$$

The composition $\pi(\psi(\cdot))$ has Lipschitz constant L_ϕ (global) on the gating output. We further use the *within-regime* gating Lipschitz $L_{\phi,\text{eq}}^{(r)}$, the maximum gating-output slope inside the regime-r interior, which is empirically 5–10$\times$ smaller than L_ϕ for tempered softmax with separated centers. This distinction matters for Theorem 1.

Per-regime predictors and the soft-gated mixture. For each regime r, a regime-specific predictor takes a NAVAR-style additive form:

$$\mu_\theta^{(r)}(z, e_u)_j = \beta_j^{(r)} + \sum_{i=1}^{K_Y} f_{ij}^{YY,(r)}(\mathcal{H}_t^{Y,(u),i}) + \sum_{p=1}^{K_X} f_{pj}^{XY,(r)}(\mathcal{H}_t^{X,(u),p}) + g_j^{(r)}(e_u), \tag{2}$$

where $f_{ij}^{YY,(r)}$ and $f_{pj}^{XY,(r)}$ are per-edge two-layer ReLU MLPs and e_u is an optional country embedding. The full regime-aware predictor is the soft-gated mixture

$$\mu_\theta(z, e_u) = \sum_{r=1}^{R} \pi_r(\psi(z)) \cdot \mu_\theta^{(r)}(z, e_u). \tag{3}$$

Each per-regime predictor satisfies joint Lipschitz conditions $\|\mu_\theta^{(r)}(\mathcal{H}_1^Y, \cdot) - \mu_\theta^{(r)}(\mathcal{H}_2^Y, \cdot)\|_2 \le L_Y^{(r)} \|\mathcal{H}_1^Y - \mathcal{H}_2^Y\|_F$, with the analogous bound on the X channel ($L_X^{(r)}$).

Linear methods provide analytic stability via spectral conditions on their VAR matrix but do not represent regime-conditional response without explicit indicators. Threshold and Markov-switching VAR represent regime structure but inherit linear dynamics within regimes and lack stability conditions for the switching system as a whole. NAVAR is nonlinear and additive but does not represent regime structure. M2 combines all three properties with the contraction bound established next.

4 Stability theorem

The architecture's stability properties decompose into three components: within-regime equilibrium dynamics, transition-step expansion at regime punctuations, and the rate at which punctuations occur. We state a rate-based contraction theorem in terms of five constants derived from the trained architecture; a global Lipschitz lemma and a stochastic error-propagation theorem (used to bound recursive predictive variance under covariate-forecast noise) appear in Appendix A, together with all proofs.

Aggregate constants. (i) $L_Y^{(r)}$, the per-regime endogenous Lipschitz of $\mu_\theta^{(r)}$ w.r.t. $\mathcal{H}^Y$, with $L_{\max} := \max_r L_Y^{(r)}$; audited post-hoc by gradient probing on held-out data, max-over-samples. (ii) $L_X^{(r)}$, analogous for the covariate channel, with $L_{X,\max} := \max_r L_X^{(r)}$. (iii) L_ϕ, the global gating Lipschitz, computed analytically from τ. (iv) $L_{\phi,\text{eq}}^{(r)}$, the within-regime gating Lipschitz. (v) $M := \max_{r,r'} \sup_z \|\mu_\theta^{(r)}(z) - \mu_\theta^{(r')}(z)\|_2$, the regime disagreement; the architecture's regime-conditional contribution beyond an additive baseline scales with M. These combine into

$$L_{\text{eq}}^{(r)} := L_Y^{(r)} + L_{\phi,\text{eq}}^{(r)} \cdot M, \qquad B_{\text{trans}} := L_{\max} + L_\phi \cdot M, \tag{4}$$

with $L_{\text{eq},\max} := \max_r L_{\text{eq}}^{(r)}$. $L_{\text{eq}}^{(r)}$ is the Lipschitz of one recursion step when the trajectory remains in regime r across consecutive lag-window updates ("equilibrium step"); B_{trans} upper-bounds any single step including those crossing regime boundaries ("transition step" or "punctuation"). A global Lipschitz lemma (Appendix A.1) shows the predictor is globally Lipschitz with constant B_{trans}, but contractivity of recursive trajectories requires the sharper equilibrium-step bound $L_{\text{eq},\max}$.

Theorem 1 (Rate-based regime-conditional contractivity)**.** *Suppose:* ***(A1)*** $L_{eq,\max} < 1$*;* ***(A2)*** $\psi \mapsto \pi(\psi)$ *is Lipschitz with constant* L_ϕ*, giving* $B_{trans} = L_{\max} + L_\phi M$*;* ***(A3)*** *the regime-transition rate* $\alpha(h) := |P(h)|/h$*, where* $P(h)$ *is the set of punctuation steps in horizon* h*, satisfies* $\alpha(h) < \alpha^* := -\log L_{eq,\max}/(\log B_{trans} - \log L_{eq,\max})$ *when* $B_{trans} > 1$*, and is unconstrained when* $B_{trans} \le 1$ *(the rate condition is vacuously satisfied). Then for every horizon* h *and every pair of initial histories,*

$$\|\mathcal{H}_h^Y - \mathcal{H}_h'^Y\|_F \;\le\; C_Y \cdot L_{eq,\max}^{(1-\alpha(h))h}\, B_{trans}^{\alpha(h)h} \cdot \|\mathcal{H}_0^Y - \mathcal{H}_0'^Y\|_F, \tag{5}$$

where C_Y *absorbs the* $\sqrt{L}$ *lag-window-Frobenius conversion. Under (A3), the per-step factor* $\rho := L_{eq,\max}^{1-\alpha} B_{trans}^{\alpha}$ *is strictly less than* 1*, and trajectories contract geometrically.*

The intuition is dwell-time accounting. Equilibrium steps contract by $L_{\text{eq}}^{(r)}$, punctuation steps expand by up to B_{trans}. Geometric contraction holds whenever the (log-)expansion contributed by punctuations is dominated by the (log-)contraction contributed by equilibrium steps, which gives the threshold α^* on punctuation frequency. The result is a switching-system analog of Hespanha–Morse average-dwell-time stability [13], specialized to neural-network-based switching dynamics with architecture-derived constants. To anchor the constants, consider two settings. On democracy indicators from the V-Dem under an unconstrained training configuration, the audit yields $\hat{L}_{\text{eq},\max} \approx 0.29$ and $\hat{B}_{\text{trans}} \approx 0.30$, both below 1. Each step is therefore contractive, and (A3) is vacuously satisfied (the trivial-contraction regime). The audited contraction rate is $\hat{\rho} \le 0.36$ across all 9 audited cells (§6.4). In contrast, on the synthetic stress test (§7), the same audit at $\lambda_M = 0$ gives $\hat{L}_{\text{eq},\max} \approx 2.38$, so the contractivity condition (A1) fails. These two cases illustrate the two regimes of the bound: the empirical panel lies within the trivial-contraction case covered by the edge-case clause, while the synthetic experiment activates the non-trivial rate condition. A trajectory-dependent sharpening (Corollary in Appendix A.3) replaces $L_{\text{eq},\max}$ with the geometric mean realized along a trajectory, tightening the bound when the trajectory spends most of its time in low-Lipschitz regimes. A companion stochastic error-propagation theorem (Appendix A.4) bounds the recursive predictive variance uniformly in horizon under independent covariate-forecast innovations.

Penalty-bridge correspondence. The training loss decomposes as $\mathcal{L} = \mathcal{L}_{\text{forecast}} + \lambda_Y \mathcal{L}_{\text{stab}} + \lambda_\phi \mathcal{L}_{\text{gate}} + \lambda_M \mathcal{L}_{\text{disagree}}$, with the three penalty components targeting $L_Y^{(r)}$, L_ϕ, and M respectively (§5). The correspondence is diagnostic, not constructive: under nonconvex training, no closed-form mapping exists from penalty coefficients to trained Lipschitz constants, and satisfaction of (A1)–(A3) is verified post-hoc by audit. This matches standard practice for spectral-norm regularization [7; 14]. The architecture's contribution is that the constants entering the contraction bound are *computable from the trained model* and *individually controllable* during training, so the failure mode (which constant violates which premise) is identifiable.

5 Training and audit

The trained model is required to satisfy (A1)–(A3) of Theorem 1 after training. Three penalty terms target the relevant constants individually: $\mathcal{L}_{\text{stab}} = \sum_r \mathbb{E}_z[\max(0, \hat{\sigma}_Y^{(r)}(z) - \tilde{L}_Y)^2]$ (soft-max-over-batch ReLU-hinged spectral norm via logsumexp at temperature 5), $\mathcal{L}_{\text{gate}} = (1/\tau)^2$, and $\mathcal{L}_{\text{disagree}} = \sum_{r \neq r'} \mathbb{E}_z \|\mu_\theta^{(r)}(z) - \mu_\theta^{(r')}(z)\|_2^2$.

We compare three enforcement modes for the within-regime $L_Y^{(r)}$: **soft penalty** ($\lambda_Y = 0.20$, target $\tilde{L}_Y = 0.55$); **projection-every-epoch** (light $\lambda_Y = 0.05$ plus hard SVD projection of each per-edge linear layer to spectral norm $\leq \sqrt{0.90}$ each epoch); and **projection-every-batch** (no soft penalty; hard SVD projection to $\leq \sqrt{0.85}$ after every optimizer step [7]). The disagreement penalty λ_M is soft in all three modes; hard projection on M is awkward (it is not a per-layer property). The gating-temperature penalty λ_ϕ is held fixed across modes.

After training, a unified audit computes empirical estimates of all constants from Theorem 1: $\hat{L}_Y^{(r)}$ via gradient probing on held-out data; $\hat{L}_X^{(r)}$ analogously; $\hat{L}_{\phi,\text{eq}}^{(r)}$ by sweeping ψ across each regime interior; $\hat{M}$ as the maximum pairwise regime-output difference on held-out data; $\hat{L}_\phi$ analytically from τ. The derived $\hat{L}_{\text{eq,max}}$, $\hat{B}_{\text{trans}}$, $\hat{\alpha}^*$, and $\hat{\rho}$ follow. We treat the bound as satisfied for a trained model when $\hat{L}_{\text{eq,max}} < 1$ at every seed in a multi-seed run. A complementary certified spectral-norm upper bound (Appendix D.2) confirms the gradient-probe estimate is non-vacuous in all our settings; the gap between the two estimators is empirically $\approx 7\%$ on V-Dem democracy indicators.

6 Empirical demonstration: democratic development (V-Dem indicators)

We demonstrate the architecture on a real-data social science forecasting task: predicting year-over-year change in clean elections across 151 countries. The aim is mechanism demonstration first and predictive benchmark second. We begin with a diagnostic that establishes the substantive empirical finding (regime-conditional signal exists in dataset's ΔY, is canceled by pooled additive predictors, and is recoverable by regime-stratified methods), then evaluate the proposed architecture against pooled neural baselines, then verify the contraction conditions.

Why V-Dem indicators is a hard test. The country-year panel of democracy indicators derived from V-Dem [8] (Appendix C) exhibits all three difficulties that motivate this work. *Multicollinearity*: indicators are expert-coded measurements of latent democratic constructs [8], with pairwise correlations exceeding 0.7 for many natural pairs and 80% of variance in the first two principal components. *Latent unit heterogeneity*: the 151 countries exhibit dynamics that grouping models identify as 5 to 9 distinct latent regimes for the income–democracy relationship alone [2]. *Regime-conditional sign changes*: predicting year-over-year change $\Delta Y_t = Y_t - Y_{t-1}$ exposes sign-changing relationships that pooled additive models cancel, while removing the trivial pooled persistence ($\rho \approx 0.98$) that would otherwise drown out any predictive contribution beyond random walk.

Data and setup. We analyze a balanced panel of country-year democracy indicators derived from V-Dem v15 [8]: 151 countries, 1970–2025. Target Y_t = `v2xel_frefair` (clean elections); covariate channel X comprises four substantively-interpretable shock-like indicators (`v2csantimv`, `v2jureform`, `v2psorgs`, `v2mecorrpt`). Lag window $L = 5$; per-country temporal split with the last 10 years held out. Final sample sizes: 6,191 training, 1,510 test observations.

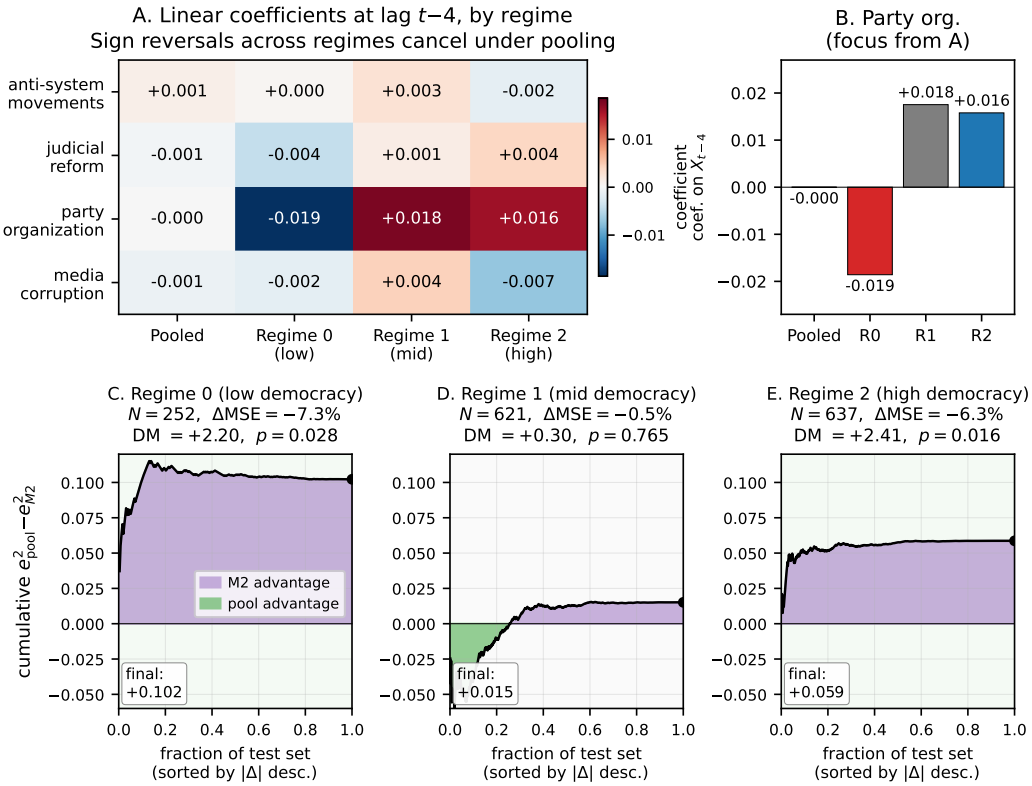


Figure 1: Diagnostic and recovery of regime-conditional signal on V-Dem ΔY. **Top (cancellation diagnostic, linear models): (A)** Linear coefficients at lag $t-4$ across pooled (column 1) and the three regime-stratified models (columns 2–4) for each of the four selected covariates. Sign reversals across regimes drive near-zero pooled coefficients. **(B)** Detail on party organization at lag $t-4$, the cleanest sign-reversal exemplar. **Bottom (recovery, M2 vs. pooled NN M1.5): (C–E)** Cumulative paired squared-error advantage across the test partition, sorted by |paired diff| descending. Final value at $x = 1$ equals $N \cdot \overline{\Delta\text{MSE}}$ (the DM test statistic's numerator). M2 advantage is concentrated in the heavy-hitters in regimes 0 and 2; in regime 1, the heavy-hitters are won by pooled before M2 catches up to a small net positive.

6.1 Diagnostic: regime-dependent signal exists and is canceled by pooling

We test directly whether regime-dependent structure is present in the data, independent of any nonlinear architecture. We define three regimes from training-set Y_{t-1} tertiles (low / mid / high democracy), and compare two linear predictors on ΔY_t: (a) a pooled linear model on the full $(Y_{t-L:t-1}, X_{t-L:t-1})$ lag window; (b) a regime-stratified linear model fit independently on each tertile, with test observations routed to their stratum. The two share identical functional form within each regime; the only difference is whether parameters are estimated jointly (pooled) or separately (stratified).

Table 1: Linear regime-stratification diagnostic on V-Dem democracy indicators (ΔY). Stratified linear M1.5 improves over pooled by $+3.02\%$ overall. All four X variables show coefficient sign reversal across regimes at one or more lag in $\{t-2, \ldots, t-5\}$.

N_{test}	Regime 0 (low) 252	Regime 1 (mid) 621	Regime 2 (high) 637	Overall 1,510
Pooled linear M1.5 (MSE)	0.00520	0.00487	0.00143	0.00347
Stratified linear M1.5 (MSE)	0.00518	0.00465	0.00140	0.00337
Improvement vs. pooled	$+0.36\%$	$+4.49\%$	$+2.01\%$	$+3.02\%$

The diagnostic establishes the substantive social science finding: *regime-conditional predictive signal exists in ΔY, is canceled by pooled additive predictors, and is recoverable by regime-stratified methods.* The cancellation mechanism is direct: four of four selected X variables exhibit coefficient sign reversal across regimes at one or more lag, so pooling averages over sign-disagreed regression coefficients and approaches their cancellation limit. Figure 1A–B visualizes this: at lag $t-4$, the pooled-linear coefficient on party organization is essentially zero (-0.000) while the per-regime coefficients are sharply opposed (-0.019 in low democracy, $+0.018$ in mid, $+0.016$ in high). The remaining question is methodological: whether a regime-aware end-to-end architecture recovers the same signal without hard partitioning of the data and while maintaining stability.

6.2 Baselines and the difficulty of forecsting a V-Dem democracy indicator

Baselines on the ΔY test partition: random walk ($\widehat{\Delta Y_t} = 0$) achieves test MSE 0.00340; pooled linear M1.5 achieves 0.00347 (*worse* than RW by -2.25%, because the high-multicollinearity, sign-canceling covariate structure inflates pooled-coefficient variance without compensating bias reduction); stratified linear M1.5 achieves 0.00337 ($+0.84\%$ over RW); and a pooled NN PANAR-X (M1.5, neural per-edge networks) achieves 0.00352 at the seed mean over 5 seeds (also worse than RW). Nonlinear flexibility alone does not exploit the regime-conditional signal; the structural assumption of regime stratification is what unlocks gain over RW. This is the regime where social science forecasting has been stuck, and where improvements of a few percent in MSE represent meaningful recovery of structure rather than easy gains.

6.3 Regime-aware architecture sweep and predictive evaluation

We sweep PANAR-X (M2) across three audited enforcement modes (soft penalty, projection-every-epoch, projection-every-batch) crossed with three values of $\lambda_M \in \{0, 0.05, 0.10\}$ and five seeds, for 45 audited runs. The architecture is compared against pooled Y-only PANAR (M1) and pooled Y+X additive PANAR-X (M1.5); M1.5 is the natural pooled additive comparison and headline baseline.

Table 2: Per-regime paired Diebold–Mariano test, M2 (mean of 5 seeds, unconstrained soft mode) vs. pooled NN M1.5. Regime 2 ($p = 0.016$) survives Holm–Bonferroni correction across $m = 3$ comparisons; regime 0 ($p = 0.028$) is nominally significant but not Holm-significant; regime 1 is statistically indistinguishable.

	N	M2 MSE	Pool MSE	DM (p)
Regime 0 (low)	252	0.00515	0.00555	$+2.20$ (0.028)
Regime 1 (mid)	621	0.00478	0.00481	$+0.30$ (0.765)
Regime 2 (high)	637	0.00136	0.00146	$+2.41$ (0.016)
Overall	1,510	0.00340	0.00352	$+2.41$ (0.016)

Predictive performance. All 9 audited cells beat pooled NN M1.5 with seed-level 95% confidence intervals strictly excluding zero; the mean $\text{M1.5}_{\text{NN}} - \text{M2}$ gap ranges from $+0.00007$ to $+0.00009$ across cells. To verify this gain at the observation level rather than the seed level, we conduct a paired Diebold–Mariano test [15] on the M2 predictions (averaged across 5 seeds at the unconstrained soft mode) vs. pooled NN M1.5 across all $1{,}510$ test observations: DM $= +2.41$, $p = 0.016$ (two-sided). The architecture's predictions have systematically lower squared error than pooled M1.5, not merely on average across seeds but on the same test observations.

Where the gain comes from. Per-regime paired DM tests (Table 2) localize the predictive contribution to the tails of the democracy index: M2 reduces MSE by 7.3% in regime 0 (low democracy) and 6.3% in regime 2 (high democracy); in regime 1 (mid) M2 and pooled are indistinguishable on the paired test. Figure 1C–E shows the cumulative paired squared-error advantage in each regime: in regimes 0 and 2 the advantage is concentrated in the heavy-hitters (largest paired differences), with M2 winning early on the sorted-by-magnitude curve and the cumulative advantage plateauing high; in regime 1 the cumulative path goes negative on the heaviest hitters before recovering to a small net positive, consistent with the DM null. Substantively: the regime-aware architecture recovers signal in the regimes where the diagnostic showed largest sign-disagreement (low and high democracies), and ties where pooled relationships happen to align with stratified ones.

6.4 Stability verification and interpretation

We verify the contraction conditions of Theorem 1 on the trained models via the audit protocol of Appendix D. All 9 audited cells satisfy $\hat{L_{\text{eq,max}}} < 1$ in $5/5$ seeds; the empirical $\hat{L_{\text{eq,max}}}$ ranges from 0.130 (most enforced) to 0.292 (unconstrained soft mode), all well within the contraction regime. The certified spectral-norm upper bound $\widetilde{L}_{\text{eq,max}}$ (Appendix D.2), which by construction dominates the gradient-probe estimate and yields a stronger guarantee, exceeds it by only $\approx 7\%$ on V-Dem indicator and remains well below 1. Combined with the empirical $\hat{\alpha} < \hat{\alpha}^*$, the implied contraction rate $\hat{\rho} \leq 0.36$ across all audited cells: recursive forecasts under the trained PANAR-X are contractive, with small input perturbations decaying over multi-step rollout.

The decomposition of Theorem 1 clarifies why the audit holds without explicit projection on V-Dem indicators: the regime-disagreement $\hat{M} \approx 0.87$ at $\lambda_M = 0$ and the per-regime $\hat{L}_Y^{(r)}$ are small enough that the gating-correction product $\hat{L}_{\phi,\text{eq}} \cdot \hat{M}$ does not push $\hat{L_{\text{eq,max}}}$ above 1. This is in contrast to the synthetic stress test of the next section, where M reaches ~ 6 and the same architecture violates the contraction condition unless a disagreement penalty is applied. The substantive content of the result is not the 3.3% overall MSE reduction over pooled NN M1.5 but the combination: regime-aware modeling provides a principled way to capture heterogeneous dynamics in social science panel data, the recovery is localized to the regimes where the diagnostic shows largest sign-disagreement, and the contraction analysis ensures that the resulting forecasts remain stable under recursive deployment.

7 Synthetic stress test: when the contraction certificate binds

The empirical democracy indicator audit holds without explicit projection because the trained M is moderate. To show the certificate is non-vacuous - that there exist data regimes where it actively constrains the architecture, and to identify what training-time intervention restores contraction when it fails - we construct a synthetic three-regime panel with deliberately strong regime contrasts and characterize the resulting stability-accuracy trade-off.

DGP and design. Univariate three-regime panel with regime determined by lagged target value: $Y_{u,t} = \beta_u + \rho_Y^{(r_t)} Y_{u,t-1} + \beta_X^{(r_t)} X_{u,t-1} + \epsilon_{u,t}$, with $\rho_Y = (0.5, 0.8, 0.7)$, $\beta_X = (0.05, 0.20, 0.40)$ (an $8\times$ range across regimes), X an independent AR(1) with autocorrelation 0.9, $U = 30$ units, $T = 50$, last 10 observations per unit held out. Empirical regime-transition rate (full panel) $\alpha \approx 0.20$. Full DGP and training details in Appendix B.

Unconstrained: predictive contribution, broken contraction. Trained without explicit Lipschitz regularization (soft penalty mode, $\lambda_M = 0$, $\lambda_Y = 0.20$, target $\tilde{L}_Y = 0.55$), M2 attains a clean predictive advantage over pooled M1.5: 5-seed mean test MSE drops from 0.1497 (M1.5) to 0.1410 (M2), gap $+0.0087$ (95% CI $[+0.0065, +0.0109]$). The audit at this optimum gives $L_{\max} = 0.77$, $L_{\phi,\text{eq}} = 0.27$, $M = 5.94$, and $L_{\text{eq,max}} = 2.38$: *the contractivity premise of Theorem 1 fails*. The violation is driven entirely by the gating-correction term $L_{\phi,\text{eq}} \cdot M = 1.6$, which dominates over $L_{\max} = 0.77 < 1$. The per-edge networks individually satisfy contraction; their disagreement is large enough to push the mixture-system bound above unity.

The Pareto frontier. Sweeping the three enforcement modes crossed with $\lambda_M \in \{0, 0.05, 0.10\}$ (45 runs total) reveals a monotonic frontier (full table in Appendix B.4; summary in Table 3): stronger enforcement reduces both $L_{\text{eq,max}}$ and the M2-vs-M1.5 gap together. Across the 45 (cell, seed) pairs, OLS regression of gap on $L_{\text{eq,max}}$ gives slope $+0.0075 \pm 0.0011$ ($R^2 = 0.54$, $p < 10^{-7}$). Unenforced cells ($\lambda_M = 0$) attain the strongest predictive contribution but fail the bound in $5/5$ seeds; full enforcement (per-batch projection at $\lambda_M = 0.10$) attains $L_{\text{eq,max}} < 1$ in $5/5$ seeds with the predictive contribution removed.

M is the lever. The decomposition $L_{\text{eq}}^{(r)} = L_Y^{(r)} + L_{\phi,\text{eq}}^{(r)} \cdot M$ makes the frontier interpretable. Under projection-every-batch with $\lambda_M = 0$, $L_Y^{(r)}$ is hard-projected to target but $L_{\text{eq,max}} \approx 2.59$ holds because $M \approx 6.95$ and $L_{\phi,\text{eq}} \cdot M \approx 1.98$. Activating $\lambda_M = 0.10$ on the same projection setting collapses M to 0.47 and $L_{\text{eq,max}}$ to 0.91 with $L_Y^{(r)}$ unchanged. *The regime-disagreement penalty is*

Table 3: Synthetic stress test: contrast at unconstrained ($\lambda_M = 0$, soft penalty) vs. most-enforced ($\lambda_M = 0.10$, projection-every-batch) training, alongside V-Dem indicators at the same configurations. "Gap" is M2 minus pooled M1.5 in test MSE units (positive favors M2); positive gap with broken bound is the failure mode the certificate identifies. Full 45-run frontier in Appendix B.4.

	Synthetic		V-Dem ind.	
	$\lambda_M = 0$	$\lambda_M = 0.10$	$\lambda_M = 0$	$\lambda_M = 0.10$
M (regime disagreement)	5.94	0.47	0.87	0.05
$L_{\text{eq,max}}$	2.38	0.91	0.29	0.13
$L_{\text{eq,max}} < 1$ in 5/5 seeds	0/5	5/5	5/5	5/5
M2 vs. M1.5 gap	+0.0087	−0.0112	+0.00009	+0.00007

essential for the bound: hard projection on L_Y alone - the conventional spectral-normalization route - does not deliver contractivity for switching systems with non-trivial regime structure. The same disagreement that drives the bound violation also drives the predictive contribution; the frontier is the locus of this trade-off, with λ_M the controllable knob.

Synthesis with V-Dem indicators. On the synthetic DGP, regime contrasts are deliberately strong, unconstrained training drives M to ≈ 5.9, and the gating-correction term pushes $L_{\text{eq,max}}$ well above the contraction threshold; λ_M is required to bring the system into the contraction regime, and reducing M also reduces predictive contribution. On the V-Dem indicator, $M \approx 0.87$ at $\lambda_M = 0$, nearly an order of magnitude smaller, and the gating-correction term is small enough that $L_{\text{eq,max}} < 1$ holds without any disagreement penalty. The two experiments together establish that the contraction certificate is necessary on data with strong sign-changing regime-conditional structure, and compatible with predictive contribution on real social science panel data with the typical heterogeneity, multicollinearity, and weak-signal structure. The architecture provides the diagnostic constants to detect which regime applies to a given dataset.

8 Limitations and conclusion

Limitations. The 3.3% overall MSE reduction over pooled NN M1.5 on V-Dem indicator (paired DM $z = 2.41$, $p = 0.016$) is small in absolute terms; we frame the result as mechanism demonstration rather than predictive benchmark. Regime structure is exogenous to the model: the regime function ψ is a fixed Lipschitz functional of state with centers set as training-data tertile midpoints, a deliberate choice to keep the contraction analysis tractable. Per-regime parameters scale linearly with regime count; with $R = 3$ this is not a practical issue for the V-Dem democracy panel, but the architecture is more parameter-hungry than pooled M1.5. The penalty-bridge correspondence is diagnostic, not constructive: practitioners must train, audit, and adjust, with the synthetic Pareto frontier providing the qualitative shape of the trade-off. Function-valued interpretability of the trained per-regime networks is inherited from the additive structure but not re-demonstrated empirically here; ICE-curve-style inspection per regime is left to future work.

Conclusion. We presented PANAR-X, a regime-aware additive forecasting architecture for heterogeneous panel time series with a verifiable contraction certificate. The contraction theorem (Theorem 1) decomposes the global Lipschitz constant into per-regime, regime-disagreement, and dwell-time components, each empirically auditable and individually controllable during training, and identifies a failure mode of standard spectral normalization on soft-gated mixture architectures. Empirical demonstration on V-Dem indicators shows that on real social science panel data, the architecture recovers regime-conditional signal canceled in pooled additive models (paired Diebold-Mariano $p = 0.016$) while the contraction certificate holds in $5/5$ seeds across all 9 audited configurations, without explicit projection. A synthetic stress test confirms the certificate is non-vacuous when regime contrasts are stronger. We do not claim large predictive gains on social science panels; we claim that regime-dependent dynamics can be modeled under formal stability constraints, that the architecture provides the diagnostic constants to detect when stability binds for a given dataset, and that the combination fills a methodological gap between pooled additive models that cancel sign-changing regime-conditional structure and unconstrained nonlinear models that risk recursive instability.

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A Full proofs

We give full proofs of the global Lipschitz lemma, the rate-based contraction theorem (Theorem 1), and a stochastic error-propagation theorem, along with two auxiliary lemmas (equilibrium-step and punctuation-step bounds).

A.1 Global Lipschitz lemma

Lemma 1 (Global Lipschitz of the soft-gated mixture)**.** *The predictor μ_θ defined in Section 3 is globally Lipschitz with constant $L_{global} = L_{\max} + L_\phi \cdot M = B_{trans}$.*

Proof. Let $z_1 = (\mathcal{H}_1^Y, \mathcal{H}^X, e)$ and $z_2 = (\mathcal{H}_2^Y, \mathcal{H}^X, e)$. Using the soft-gating mixture (3), write

$$\begin{aligned}\mu_\theta(z_1) - \mu_\theta(z_2) = &\underbrace{\sum_r \big[\pi_r(\psi(z_1)) - \pi_r(\psi(z_2))\big]\mu_\theta^{(r)}(z_1)}_{\text{(I): gating-driven}} \\ &+ \underbrace{\sum_r \pi_r(\psi(z_2))\big[\mu_\theta^{(r)}(z_1) - \mu_\theta^{(r)}(z_2)\big]}_{\text{(II): within-regime}}\end{aligned} \tag{6}$$

The decomposition is exact: add and subtract $\sum_r \pi_r(\psi(z_2))\mu_\theta^{(r)}(z_1)$.

Bound on (I). Since $\sum_r \pi_r(s) = 1$ for all s, $\sum_r[\pi_r(\psi(z_1)) - \pi_r(\psi(z_2))] = 0$, so we can subtract any reference $\mu_\theta^{(r^*)}(z_1)$ from each summand:

$$\begin{aligned}\|(\mathrm{I})\|_2 &\le \sum_r |\pi_r(\psi(z_1)) - \pi_r(\psi(z_2))| \cdot \|\mu_\theta^{(r)}(z_1) - \mu_\theta^{(r^*)}(z_1)\|_2 \\ &\le M \cdot \sum_r |\pi_r(\psi(z_1)) - \pi_r(\psi(z_2))| \\ &\le M \cdot L_\phi \cdot |\psi(z_1) - \psi(z_2)| \\ &\le L_\phi M \cdot \|\mathcal{H}_1^Y - \mathcal{H}_2^Y\|_F.\end{aligned}$$

Bound on (II). A convex combination of per-regime differences:

$$\|(\mathrm{II})\|_2 \le \sum_r \pi_r(\psi(z_2)) \cdot L_Y^{(r)} \|\mathcal{H}_1^Y - \mathcal{H}_2^Y\|_F \le L_{\max} \|\mathcal{H}_1^Y - \mathcal{H}_2^Y\|_F.$$

Combining. By the triangle inequality applied to (6), $\|\mu_\theta(z_1) - \mu_\theta(z_2)\|_2 \le (L_{\max} + L_\phi M)\|\mathcal{H}_1^Y - \mathcal{H}_2^Y\|_F$. The covariate-channel statement follows analogously. □

A.2 Auxiliary lemmas: equilibrium and punctuation step bounds

For the proof of Theorem 1 we need two refined per-step bounds. Consider two recursive trajectories $\{z_j\}_{j=0}^h$ and $\{z'_j\}_{j=0}^h$ updated by $\mathcal{H}_{j+1}^Y = (\mathcal{H}_j^Y[1:], \mu_\theta(z_j))$, $\mathcal{H}_{j+1}^X = (\mathcal{H}_j^X[1:], X_{j+1})$ with the same exogenous covariate sequence. Define the regime indicator $r_j := \arg\max_r \pi_r(\psi(z_j))$, the equilibrium set $E(h) := \{j : r_j = r_{j-1}$ on both trajectories$\}$, the punctuation set $P(h) := \{j : r_j \neq r_{j-1}$ on at least one trajectory$\}$, and $\delta_j := \mathcal{H}_j^Y - \mathcal{H}_j'^Y$.

Lemma 2 (Equilibrium-step Lipschitz)**.** *If step $i \in E(h)$, then $\|\mu_\theta(z_i) - \mu_\theta(z'_i)\|_2 \le L_{eq}^{(r_i)} \|\delta_i\|_F$ where $L_{eq}^{(r_i)} := L_Y^{(r_i)} + L_{\phi,eq}^{(r_i)} M$.*

Proof. Both z_i and z'_i lie in $\mathcal{Z}^{(r_i)}$, so apply the decomposition (6) restricted to the regime interior. Term (I) uses the within-regime gating Lipschitz $L_{\phi,\mathrm{eq}}^{(r_i)}$ rather than the global L_ϕ: $\|(\mathrm{I})\|_2 \le L_{\phi,\mathrm{eq}}^{(r_i)} M \|\delta_i\|_F$. Term (II) uses $L_Y^{(r_i)}$ since the convex combination $\sum_r \pi_r(\psi(z'_i))$ concentrates near r_i on the regime interior: $\|(\mathrm{II})\|_2 \le L_Y^{(r_i)} \|\delta_i\|_F$. Triangle inequality. □

Lemma 3 (Punctuation-step Lipschitz)**.** *If step $i \in P(h)$, then $\|\mu_\theta(z_i) - \mu_\theta(z'_i)\|_2 \le B_{trans} \|\delta_i\|_F$.*

Proof. At a punctuation step, the trajectories may straddle a regime boundary and the within-regime restriction does not apply. Lemma 1 gives the global Lipschitz bound directly. □

A.3 Proof of Theorem 1

Step 1: per-step bound. Apply Lemma 2 if $j \in E(h)$ and Lemma 3 if $j \in P(h)$: $\|\mu_\theta(z_j) - \mu_\theta(z'_j)\|_2 \le A_j \|\delta_j\|_F$ with $A_j = L_{\text{eq}}^{(r_j)}$ on equilibrium steps and $A_j = B_{\text{trans}}$ on punctuation steps.

Step 2: multi-step accumulation. The history shift gives δ_{j+1} as a lag-shift of δ_j with one new entry equal to $\mu_\theta(z_j) - \mu_\theta(z'_j)$. After L consecutive steps (one full lag-window refresh), the accumulated $\|\delta_{j+L}\|_F$ is bounded by $\sqrt{L} \cdot \max_{i=j,\dots,j+L-1} \|\mu_\theta(z_i) - \mu_\theta(z'_i)\|_2$. Iterating across L-step blocks: $\|\delta_h\|_F \le C_Y \cdot \prod_{j=0}^{h-1} A_j \cdot \|\delta_0\|_F$, where C_Y absorbs the $\sqrt{L}$ lag-window-Frobenius conversion.

Step 3: bounding the product. Conservatively bound all equilibrium steps by $L_{\text{eq,max}}$ and all punctuation steps by B_{trans}: $\prod_j A_j \le L_{\text{eq,max}}^{|E(h)|} \cdot B_{\text{trans}}^{|P(h)|}$. With $|E(h)| = (1-\alpha(h))h$ and $|P(h)| = \alpha(h)h$, we obtain $\|\delta_h\|_F \le C_Y \cdot L_{\text{eq,max}}^{(1-\alpha(h))h} B_{\text{trans}}^{\alpha(h)h} \|\delta_0\|_F$.

Step 4: contraction under the rate condition. The bound contracts geometrically iff $L_{\text{eq,max}}^{1-\alpha} B_{\text{trans}}^{\alpha} < 1$, i.e., $(1-\alpha)\log L_{\text{eq,max}} + \alpha \log B_{\text{trans}} < 0$, equivalently $\alpha < (-\log L_{\text{eq,max}})/(\log B_{\text{trans}} - \log L_{\text{eq,max}}) = \alpha^*$, which is assumption (A3). □

Corollary 1 (Trajectory-dependent contraction rate)**.** *Let* $\bar{L}_Y(\{r_j\}) := \exp\big(\frac{1}{|E(h)|}\sum_{j\in E(h)} \log L_{eq}^{(r_j)}\big)$ *be the equilibrium geometric mean realized along a trajectory's regime sequence. Then the bound of Theorem 1 holds with* $L_{eq,\max}$ *replaced by* $\bar{L}_Y(\{r_j\})$*, sharpening when the trajectory spends most of its time in low-Lipschitz regimes.*

Proof. Apply the proof of Theorem 1 but bound each equilibrium-step factor by its own $L_{\text{eq}}^{(r_j)}$ rather than $L_{\text{eq,max}}$. The product of equilibrium-step factors becomes $\prod_{j\in E(h)} L_{\text{eq}}^{(r_j)} = \bar{L}_Y(\{r_j\})^{|E(h)|}$ by definition of the geometric mean. □

A.4 Stochastic error propagation

Theorem 2 (Regime-aware stochastic error propagation)**.** *Under (A1)–(A3) of Theorem 1 and additionally* ***(A4)*** *covariate-forecast innovations* $\{\Delta X_k\}$ *are mutually independent with* $\mathbb{E}\|\Delta X_k\|_F^2 \le \tau_X^2$*, the recursive predictive variance is bounded uniformly in horizon:*

$$\mathbb{E}\|\mathcal{H}_h^Y - \mathcal{H}_h'^Y\|_F^2 \le C_Y^2 \rho^{2h} \|\mathcal{H}_0^Y - \mathcal{H}_0'^Y\|_F^2 + \frac{C_Y^2 L_{X,\max}^2 \tau_X^2}{1-\rho^2}, \tag{7}$$

where $\rho < 1$ *is from Theorem 1.*

Proof. The recursion under stochastic covariate-forecast innovations is non-homogeneous: $\delta_{j+1}^Y = A_j \delta_j^Y + B_j \Delta X_j$, where A_j is the step-j Y-channel Jacobian (regime-dependent: $L_{\text{eq}}^{(r_j)}$ in equilibrium, bounded by B_{trans} at punctuation) and B_j is the step-j X-channel Jacobian (bounded by $L_X^{(r_j)} \le L_{X,\max}$). The closed-form solution is $\delta_h^Y = \big(\prod_{i=0}^{h-1} A_i\big)\delta_0^Y + \sum_{k=0}^{h-1} \big(\prod_{i=k+1}^{h-1} A_i\big) B_k \Delta X_k$. Take expectations of the squared Frobenius norm. By (A4), the $\{\Delta X_k\}$ are mutually independent, so cross-product terms vanish in the second moment:

$$\mathbb{E}\|\delta_h^Y\|_F^2 \le \big\|\prod_{i=0}^{h-1} A_i\big\|_2^2 \|\delta_0\|_F^2 + \sum_{k=0}^{h-1} \big\|\prod_{i=k+1}^{h-1} A_i\big\|_2^2 \|B_k\|_2^2 \mathbb{E}\|\Delta X_k\|_F^2.$$

The first term is bounded by Theorem 1: $\|\prod_i A_i\|_2 \le C_Y \rho^h$, giving $C_Y^2 \rho^{2h} \|\delta_0\|_F^2$. For the second term, apply Theorem 1 to the sub-trajectory from $k+1$ to $h-1$: $\|\prod_{i=k+1}^{h-1} A_i\|_2 \le C_Y \rho^{h-k-1}$. Combined with $\|B_k\|_2 \le L_{X,\max}$ and $\mathbb{E}\|\Delta X_k\|_F^2 \le \tau_X^2$: $\sum_{k=0}^{h-1} C_Y^2 \rho^{2(h-k-1)} L_{X,\max}^2 \tau_X^2 = C_Y^2 L_{X,\max}^2 \tau_X^2 \sum_{j=0}^{h-1} \rho^{2j} \le C_Y^2 L_{X,\max}^2 \tau_X^2/(1-\rho^2)$ where the geometric sum converges because $\rho < 1$ under (A3). □

The first term is the contraction of initial-condition error; the second is the bounded steady-state variance from covariate-forecast noise accumulation. The bound diverges as $\alpha(h) \to \alpha^*$ ($\rho \to 1$): at the rate-condition limit, variance accumulates without bound. Setting $\alpha = 0$ recovers the additive-baseline NAVAR error-propagation result with $\rho = L_Y$.

B Synthetic data and full Pareto frontier

B.1 DGP parameters

The synthetic panel of §7 is generated as follows. For each unit $u \in \{1, \ldots, 30\}$ and time $t \in \{1, \ldots, 50\}$: per-unit intercept $\beta_u \sim \mathcal{N}(0, 0.5^2)$, sampled once per unit; initial values $Y_{u,0}, X_{u,0} \sim \mathcal{N}(0, 0.3^2)$; covariate evolution $X_{u,t} = 0.9X_{u,t-1} + \nu_{u,t}$, $\nu_{u,t} \sim \mathcal{N}(0, 0.5^2)$; target evolution: regime r_t from $Y_{u,t-1}$ via thresholds $c_1 = -0.6, c_2 = 0.6$; then $Y_{u,t} = \beta_u + \rho_Y^{(r_t)} Y_{u,t-1} + \beta_X^{(r_t)} X_{u,t-1} + \epsilon_{u,t}$ with $(\rho_Y^{(1)}, \rho_Y^{(2)}, \rho_Y^{(3)}) = (0.5, 0.8, 0.7)$, $(\beta_X^{(1)}, \beta_X^{(2)}, \beta_X^{(3)}) = (0.05, 0.20, 0.40)$, and $\epsilon_{u,t} \sim \mathcal{N}(0, 0.3^2)$. The data-generation seed is fixed (`np.random.default_rng(42)`); only model-initialization seeds vary across the 5 replications per cell. On the generated training data, regime occupancy is $28.2\%/37.5\%/34.2\%$ (low/mid/high) and the empirical regime-transition rate is $\alpha_{\text{train}} = 0.217$ ($\alpha = 0.204$ on the full panel; the model audit reports $\alpha_{\text{test}} = 0.156$).

B.2 Training hyperparameters

All 45 sweep runs use Adam with learning rate 3×10^{-3}, weight decay 10^{-4}, batch size 128, and 300 training epochs. Per-edge networks are two-layer ReLU MLPs with hidden width 32. The gating softmax temperature is initialized at $\tau_{\text{init}} = 0.5$. Gating-temperature regularization $\lambda_\phi = 0.005$. Regime centers are fixed at training-data $1/6, 1/2, 5/6$ quantiles of $\psi(z) = z[L][0]$.

B.3 Predictive contribution at the unconstrained optimum

Table 4: Test MSE at the soft-penalty unconstrained ($\lambda_M = 0$) training configuration, 5-seed mean. Decomposition isolates contributions from nonlinear X channel (M1.5 vs. M1) and regime structure (M2 vs. M1.5).

Predictor	Test MSE
RW (random walk)	0.1893
AR(1) + intercept	0.1895
M1 (Y-only PANAR)	0.1895
M1.5 (additive PANAR-X, no regimes)	0.1497
M2 (regime-aware PANAR-X), 5-seed mean	0.1410
Gain decomposition:	
Nonlinear X channel (M1.5 vs. M1)	$+0.0398$ ($+20.98\%$)
Regime structure (M2 vs. M1.5), 5-seed mean	$+0.0087$ ($+5.82\%$)
seed-level 95% CI on gap	$[+0.0065, +0.0109]$

B.4 Full Pareto frontier across 45 training runs

The frontier is monotonic across all 9 cells: stronger enforcement (higher λ_M, transition from soft to per-epoch to per-batch projection) reduces both $L_{\text{eq,max}}$ and the M2-vs-M1.5 gap together. Three regimes are visible: unenforced cells ($\lambda_M = 0$) attain the strongest predictive contribution but fail the contractivity premise in $5/5$ seeds; intermediate enforcement ($\lambda_M = 0.05$, soft penalty) sits on the boundary with a CI-significant predictive gap and partial satisfaction of the bound; full enforcement (per-epoch or per-batch projection at $\lambda_M = 0.10$) attains $L_{\text{eq,max}} < 1$ in $5/5$ seeds, with the predictive contribution removed.

Table 5: Full Pareto frontier across enforcement modes and disagreement-penalty strengths ($n = 5$ seeds per cell, 5-seed means). Gap = M1.5 MSE − M2 MSE; positive means M2 better. $L_{\phi,\text{eq}} \cdot M$ is the gating-correction contribution to the contractivity bound from Theorem 1.

Mode	λ_M	$L_{\text{eq,max}}$	M	$L_{\phi,\text{eq}} \cdot M$	gap (95% t-CI)	$L_{\text{eq,max}}<1$
soft penalty	0.00	2.38	5.94	1.62	$+0.0087$ $[+0.0065, +0.0109]$	0/5
	0.05	1.02	0.70	0.19	$+0.0046$ $[+0.0015, +0.0078]$	2/5
	0.10	0.96	0.42	0.11	$+0.0028$ $[+0.0003, +0.0054]$	3/5
proj. per epoch	0.00	2.56	6.79	1.93	$+0.0071$ $[+0.0067, +0.0076]$	0/5
	0.05	1.02	0.71	0.19	-0.0049 $[-0.0076, -0.0022]$	2/5
	0.10	0.92	0.44	0.12	-0.0073 $[-0.0097, -0.0048]$	**5/5**
proj. per batch	0.00	2.59	6.95	1.98	$+0.0066$ $[+0.0058, +0.0075]$	0/5
	0.05	0.99	0.74	0.20	-0.0089 $[-0.0122, -0.0055]$	2/5
	0.10	0.91	0.47	0.12	-0.0112 $[-0.0142, -0.0083]$	**5/5**

C V-Dem data and preprocessing

C.1 Variable selection

The target Y_t = `v2xel_frefair` (V-Dem "Clean Elections" index) is a high-level democracy indicator constructed by V-Dem from expert-coded sub-indicators via Bayesian item-response models [8]. Range $[0, 1]$; standard deviation ~ 0.30 pooled, ~ 0.06 in first differences.

The four covariates were selected from the V-Dem indicator set under two criteria: (i) shock-like temporal variation (the fraction of country-years with $|\Delta X_t| > 0.05$ should be of similar magnitude to that of ΔY_t); and (ii) lagged correlation with ΔY above noise floor. `v2csantimv` (anti-system movement strength), `v2jureform` (judicial reforms in the past year), `v2psorgs` (party organization), and `v2mecorrpt` (media corruption / capture). These are V-Dem-internal expert-coded indicators with documented substantive interpretations and are independent of `v2xel_frefair` in the V-Dem index aggregation.

C.2 Sample construction and split

We use the V-Dem v15 country-year panel restricted to countries with complete observation series across all selected variables in the 1970–2025 window. After this restriction, the panel is balanced at 151 countries × 56 years.

The lag window is $L = 5$. For each country, we hold out the last 10 years for testing; earlier years are training. Per-country split ensures every country contributes both training and test observations, which matters because country-level effects are absorbed into per-unit intercepts during training. Final sample sizes after lag-window construction: 6,191 training observations, 1,510 test observations.

C.3 Regime boundaries

Regime centers c_r and regime boundaries b_1, b_2 are computed from training-set Y_{t-1} tertiles. Empirical values: $(c_1, c_2, c_3) = (0.003, 0.41, 0.911)$, $(b_1, b_2) = (0.168, 0.674)$. The regime function is fixed by these training-derived constants and is not retrained on test data.

X normalization in the diagnostic. The linear coefficients plotted in Figure 1A–B are fit on *unstandardized* covariates, so coefficient magnitudes are directly comparable across regimes. The companion diagnostic script that prints sign-disagreement counts standardizes X using training-set mean and standard deviation; the sign-reversal pattern is identical in both, but coefficient magnitudes differ (party-organization at lag $t-4$: unstandardized $-0.019/+0.018/+0.016$ vs. standardized $-0.028/+0.027/+0.024$). Either presentation supports the four-of-four sign-reversal claim.

C.4 Target variable: ΔY vs. Y levels

We forecast $\Delta Y_t = Y_t - Y_{t-1}$ rather than Y_t directly. Pooled persistence in Y is $\rho \approx 0.98$, meaning the random-walk baseline ($\hat{Y}_t = Y_{t-1}$) achieves test MSE of order 10^{-3} on Y-levels, against unconditional variance ~ 0.09. Predicting ΔY removes this trivial persistence and re-anchors the random-walk baseline to "predict zero change," which on V-Dem test data has MSE 0.00340. Under this baseline, beating RW skill becomes a meaningful test of the architecture's contribution beyond persistence.

D Audit protocol

We describe two complementary estimators of the Theorem 1 constants on trained models: a gradient-probe sample estimator (§D.1) used as the headline audit number throughout the empirical sections, and a certified spectral-norm upper bound (§D.2) that bounds the gradient-probe estimate from above. Both are deterministic given trained model weights. The certified bound is conservative; the gradient-probe estimate is a sample maximum and may underestimate the true Lipschitz if the empirical batch does not realize the worst-case Jacobian.

D.1 Gradient-probe sample estimator

Per-regime endogenous Lipschitz $\hat{L}_Y^{(r)}$. For each regime r and a held-out batch of 200 samples, compute the per-sample Y-channel Jacobian $J^{Y,(r)}(z) := \partial \mu_\theta^{(r)}(z)/\partial \mathcal{H}^Y$ via automatic differentiation, take the Frobenius norm $\|J^{Y,(r)}(z)\|_F$, and report the maximum over the batch: $\hat{L}_Y^{(r)} = \max_z \|J^{Y,(r)}(z)\|_F$.

Per-regime covariate Lipschitz $\hat{L}_X^{(r)}$. Analogously, with $J^{X,(r)}(z) := \partial \mu_\theta^{(r)}(z)/\partial \mathcal{H}^X$.

Global gating Lipschitz $\hat{L}_\phi$. Sweep ψ across the full empirical range $[\psi_{\min}, \psi_{\max}]$ on a 500-point grid, evaluate $\pi(\psi)$ at each point, and report $\hat{L}_\phi = \max_i \|\pi(\psi_{i+1}) - \pi(\psi_i)\|_1 / |\psi_{i+1} - \psi_i|$.

Within-regime gating Lipschitz $\hat{L}_{\phi,\text{eq}}^{(r)}$. For each regime r, restrict to test observations with regime indicator $= r$, sweep ψ on a 50-point grid covering the regime's empirical 25th–75th percentile range, and apply the same finite-difference estimator as $\hat{L}_\phi$.

Regime disagreement $\hat{M}$. For all pairs $r \neq r'$ and all held-out test observations, compute $|\mu_\theta^{(r)}(z) - \mu_\theta^{(r')}(z)|$, take the maximum over test points, then maximum over regime pairs.

Derived constants. $\hat{L}_{\text{eq,max}} = \max_r[\hat{L}_Y^{(r)} + \hat{L}_{\phi,\text{eq}}^{(r)} \hat{M}]$; $\hat{B}_{\text{trans}} = \max_r \hat{L}_Y^{(r)} + \hat{L}_\phi \hat{M}$; $\hat{\alpha}^* = (-\log \hat{L}_{\text{eq,max}})/(\log \hat{B}_{\text{trans}} - \log \hat{L}_{\text{eq,max}})$ when $\hat{L}_{\text{eq,max}} < 1 < \hat{B}_{\text{trans}}$. When $\hat{B}_{\text{trans}} \leq 1$, the rate condition (A3) is vacuously satisfied and $\hat{\alpha}^*$ is reported as ∞ (no constraint on the empirical punctuation rate).

Empirical punctuation rate $\hat{\alpha}$. For each unit u in the test set, compute the predicted regime sequence $\{\hat{r}_j^{(u)}\}$ via $\arg\max_r \pi_r(\psi(z))$, count transitions, and average the per-unit transition rate.

Empirical contraction rate $\hat{\rho}$. $\hat{\rho} := \hat{L}_{\text{eq,max}}^{1-\hat{\alpha}} \hat{B}_{\text{trans}}^{\hat{\alpha}}$. We treat the trained model as certifiably contractive when $\hat{L}_{\text{eq,max}} < 1$ and either $\hat{B}_{\text{trans}} \leq 1$ (in which case $\hat{\rho} \leq \hat{L}_{\text{eq,max}} < 1$ for all $\hat{\alpha}$) or $\hat{\alpha} < \hat{\alpha}^*$ (in which case $\hat{\rho} < 1$ by Theorem 1).

D.2 Certified spectral-norm upper bound

The per-edge functions in PANAR-X are two-layer ReLU MLPs with weight matrices $W_1^{(r,e)} \in \mathbb{R}^{h \times d}$ (input-to-hidden) and $W_2^{(r,e)} \in \mathbb{R}^{1 \times h}$ (hidden-to-output) for each regime r and edge e. Since ReLU

is 1-Lipschitz, the Lipschitz constant of each per-edge function $f^{(r,e)} : \mathbb{R}^d \to \mathbb{R}$ admits the analytical upper bound $\mathrm{Lip}(f^{(r,e)}) \le \sigma(W_1^{(r,e)}) \cdot \sigma(W_2^{(r,e)})$, where $\sigma(\cdot)$ denotes the spectral (ℓ_2 operator) norm. This is a pointwise bound: the inequality holds for all inputs, not only for the held-out batch. Spectral norms are computed by power iteration on $W_1^{(r,e)}(W_1^{(r,e)})^\top$ and analytically for the output layer.

The per-regime endogenous certified bound is the sum over Y-channel edges: $\widetilde{L}_Y^{(r)} := \sum_{e \in E_Y} \sigma(W_1^{(r,e)}) \cdot \sigma(W_2^{(r,e)})$ for the additive predictor. The certified equilibrium-step bound is $\widetilde{L}_{\text{eq,max}} = \max_r [\widetilde{L}_Y^{(r)} + \hat{L}_{\phi,\text{eq}}^{(r)} \hat{M}]$, replacing the gradient-probe $\hat{L}_Y^{(r)}$ with the certified $\widetilde{L}_Y^{(r)}$.

By construction $\widetilde{L}_Y^{(r)} \ge \hat{L}_Y^{(r)}$ for all regimes, so $\widetilde{L}_{\text{eq,max}} < 1$ implies $\hat{L}_{\text{eq,max}} < 1$ and provides a stronger guarantee.

Empirical comparison on V-Dem. 3 seeds, vanilla MSE training, 100 epochs. Per-regime L_Y values from the same trained model:

	Probe ($\hat{L}_Y^{(r)}$)	Certified ($\widetilde{L}_Y^{(r)}$)	Gap (cert−probe)
Regime 0	0.129 ± 0.011	0.136 ± 0.014	$+0.007$
Regime 1	0.152 ± 0.022	0.157 ± 0.023	$+0.004$
Regime 2	0.024 ± 0.015	0.030 ± 0.012	$+0.006$
Max over regimes	0.155 ± 0.019	0.164 ± 0.014	$+0.009$ ($+7\%$)

The certified upper bound exceeds the gradient-probe sample maximum in all 9 regime-seed cells, confirming the bound is non-vacuous, and the gap is small ($\approx 7\%$ in absolute terms at the maximum-regime level). Both $\hat{L}_{\text{eq,max}} \ll 1$ and $\widetilde{L}_{\text{eq,max}} \ll 1$ hold on V-Dem, so the contraction conclusion of Theorem 1 holds whether we use the conservative certified bound or the gradient-probe estimate.

E Hyperparameter sensitivity and reproducibility

Gating temperature $\tau_{\text{init}} = 0.5$. Earlier configurations used $\tau_{\text{init}} = 3.0$, producing soft gating with broad π_r overlap that empirically averaged across regimes. A diagnostic showed that hard tertile stratification recovered regime-conditional signal that soft gating with broad τ did not. Setting $\tau_{\text{init}} = 0.5$ (sharper gating) and $\lambda_\phi = 0.005$ (lower temperature regularization, keeping gating sharp) recovered this signal under the soft-gating architecture. We did not do a formal sensitivity sweep on τ_{init}.

Stratified-baseline training epochs. The stratified-NN baseline (a separate M1.5 PANAR-X trained per regime) is trained for 900 epochs per regime, $3\times$ the 300 epochs of the pooled NN baseline, to roughly compensate for the smaller per-regime training sample (about $1/3$ of the pooled sample). Validation-based early stopping was deferred. The stratified NN baseline serves as a diagnostic of NN-based regime-stratification overfitting on small per-regime samples.

Spectral-norm targets. The soft penalty target $\tilde{L}_Y = 0.55$, projection-every-epoch target 0.90 (with light soft penalty target 0.85), and projection-every-batch target 0.85 are conservative choices ensuring the trained $L_Y^{(r)}$ stays below the unit-Lipschitz threshold by a margin. Tighter targets (e.g., 0.4) have been tried in synthetic preliminaries and produce qualitatively similar Pareto behavior with shifted absolute MSE values.

Reproducibility. Random seeds are set via `torch.manual_seed` and `np.random.seed` immediately before each model instantiation; data generation uses `np.random.default_rng(42)` (synthetic) or fixed CSV input (V-Dem). Audit functions are deterministic given trained weights. Test MSE values reproduce on independent T4 GPU hardware to within $\sim 10^{-5}$; audit constants based on max-over-samples Jacobian probes drift by up to $\sim 10^{-3}$ across hardware due to non-deterministic CUDA matmul kernels, but every substantive claim in the paper (paired-DM significance, the contraction certificate holding in $5/5$ seeds across all 9 audited cells, the sign-reversal diagnostic, and the synthetic stress-test ordering) reproduces independent of hardware.

The reproducibility package supports two reproduction paths. The audit-only path executes 141 numerical checks against archived artifact pickles in under 30 seconds on any machine, with no GPU required, and reproduces every numerical claim in the paper. Full retraining on a single T4 GPU takes approximately 3.5 hours: ~ 1 minute for synthetic baselines, ~ 17 minutes for the synthetic Pareto sweep, ~ 175 minutes for the V-Dem sweep (the dominant cost), ~ 23 minutes for postanalysis, ~ 2 minutes for the certified-bound audit, and ~ 15 seconds for figure generation. The V-Dem sweep dominates because its per-batch Jacobian penalty, required for the contraction certificate, scales with training-set size ($N_{\text{train}} = 6{,}191$ vs. $1{,}050$ for the synthetic panel). CPU-only retraining is approximately 12 CPU-hours and is supported but not recommended for reviewer use. Total runtime exceeds the free-tier Colab idle timeout (90 minutes); reviewers using Colab should use Colab Pro or run the audit-only path.

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